

Wilson Harper

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wilsonharper.net

Houston, TX

Education

Rice University	Class of 2029
• B.S. Electrical & Computer Engineering; Minor in Entrepreneurship • Elected as New Student Representative in Martel College Parliament	
Davis High School	2021 - 2025
• 3.99 GPA and IB Diploma, earned Washington State Seal of Biliteracy in Spanish	

Engineering Experience

Incoming Embedded Engineering Intern, RTX BBN Technologies	2026
• Will prototype embedded multi-domain systems and PNT algorithms for harsh environments	
Undergraduate Researcher, Rice Experimental Pixels Lab	2026 - Present
• Selected to join the Rice EXP Lab as an undergraduate researcher • Researching a compact and directional wind system to build realism in extended reality environments	
Active Fin Stabilization, Rice Eclipse Rocketry	2025 - Present
• Designing avionics for reliable rocket active fin stabilization system • Integrating RP2350 and I2C/UART sensors on custom PCB to control servo-actuated fins • Active fin stabilization system will go on space-shot rocket to exceed the Kármán line	
Thrust Stand Design, Rice Flight	2025 - 2026
• Led design of Arduino thrust test stand to measure efficiency for UAV optimization • Built Pi-powered ADS-B aircraft tracker with custom ground plane antenna and display system	
UAV Team Captain, Technology Student Association	2018 - 2025
• Led design of position-holding UAV in GPS-denied environments using optical flow and LiDAR • Designed a modular 3D-printed airframe and servo-actuated gripper to move payloads • Implemented EKF sensor fusion, remote telemetry, and power distribution systems	
Researcher, MIT Lincoln Laboratory RISE	2024
• One of ~4% of applicants selected for DoD-sponsored free research experience • Researched, built, and programmed Doppler and synthetic aperture radars, mentored by Lincoln Labs staff	

Professional Experience

Teaching Assistant, BUSI 361: Communication for Entrepreneurs	2026
• Facilitating instruction on communication, conflict resolution, networking, and workplace dynamics • Coaching students on public speaking to effectively communicate complex concepts	
Summer Camp Staff, Spokane, WA	2025
• Responsible for up to 65 guests at remote location accessible only by boat • Learned high-stakes logistics planning due to remote environment • Duties ranged from comforting homesick children to cleaning large amounts of septic waste	

Skills

- **Software:** Python, CAD (Fusion, OnShape), avionics, ArduPilot, MS Office (Word, Excel, Powerpoint)
- **Hardware:** LiDAR, radar, telemetry, sensor integration, UAVs
- **Tools:** soldering, 3D printing, CNC routing, hand tools, multimeters, breadboards